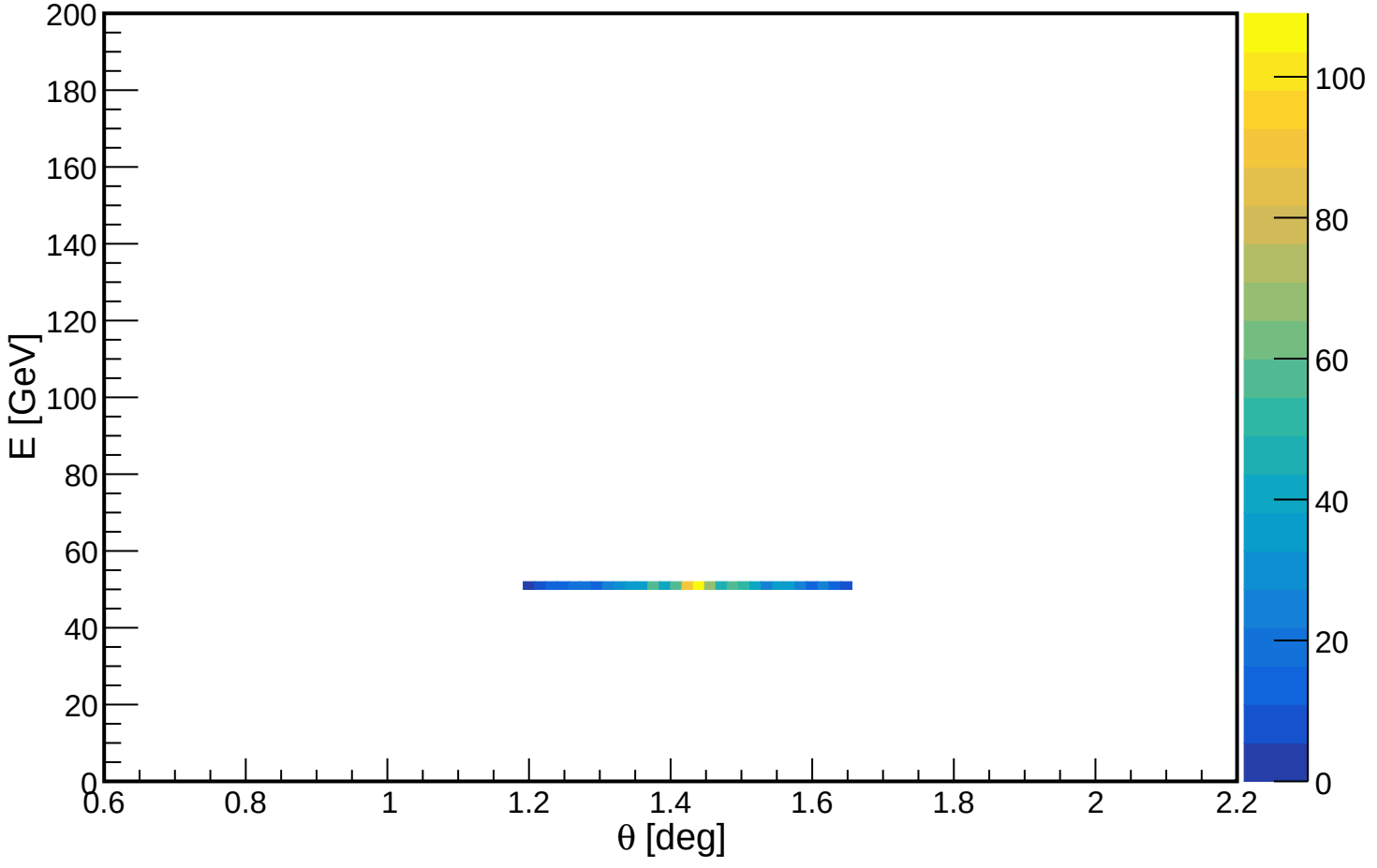
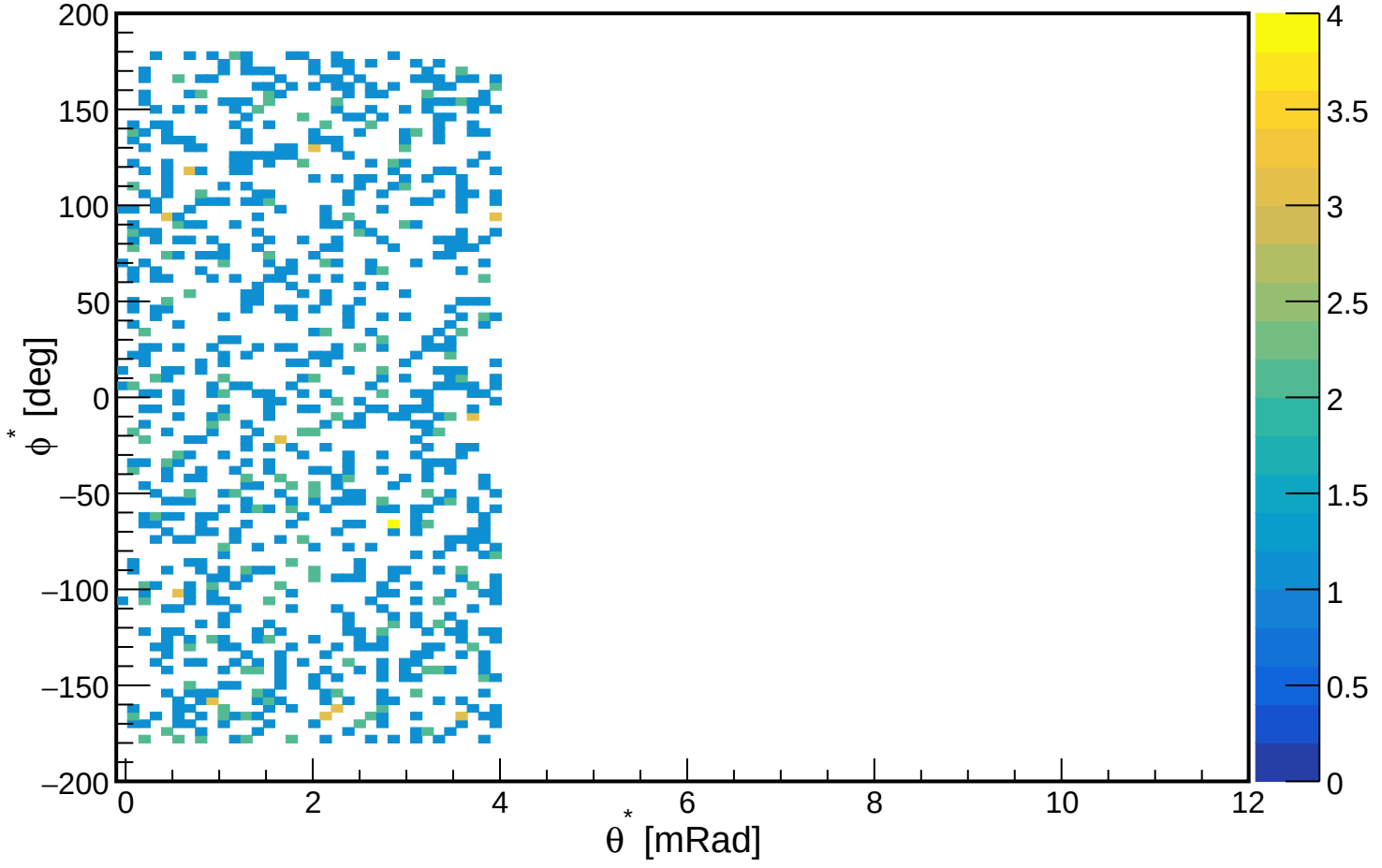
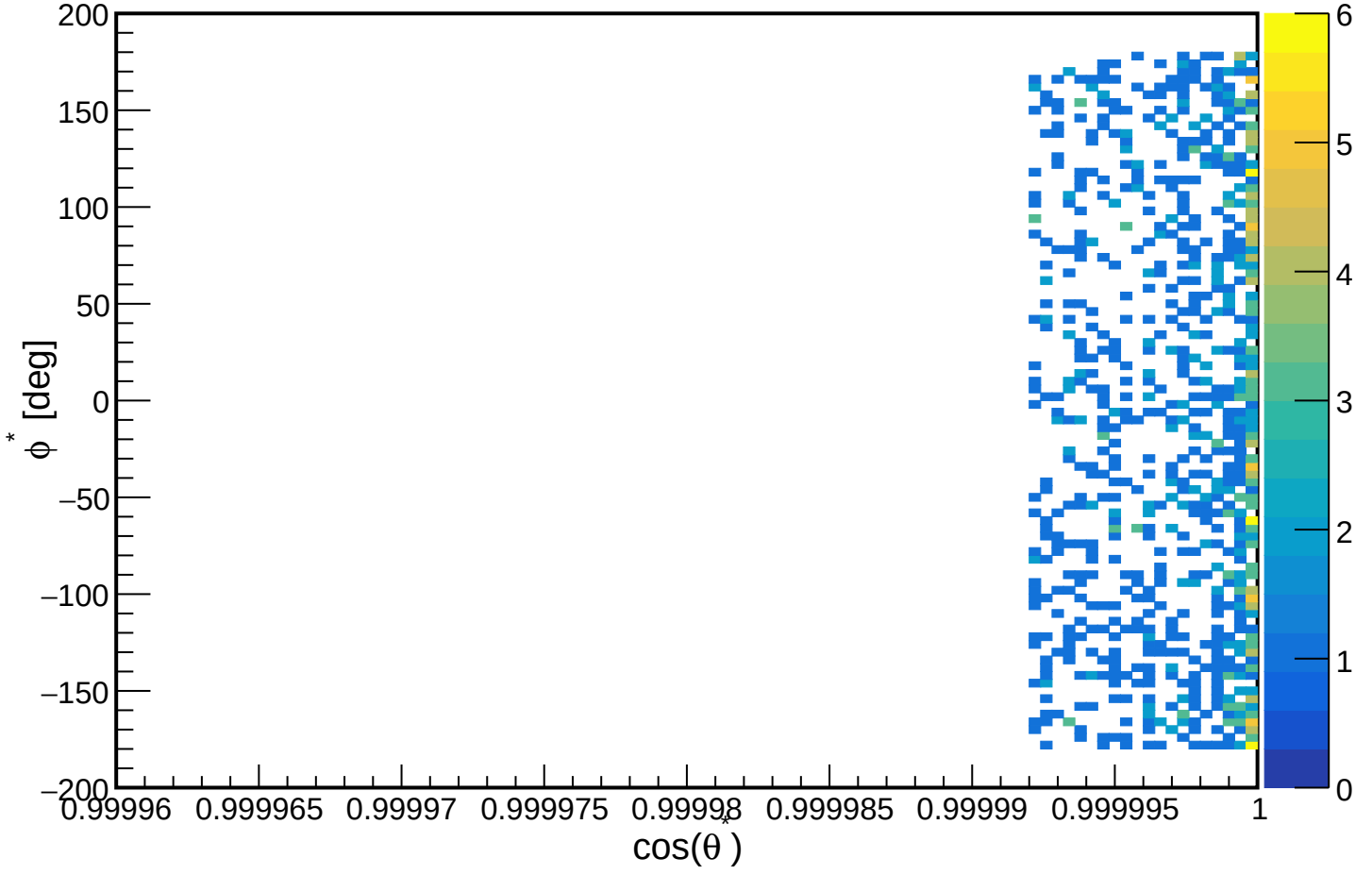
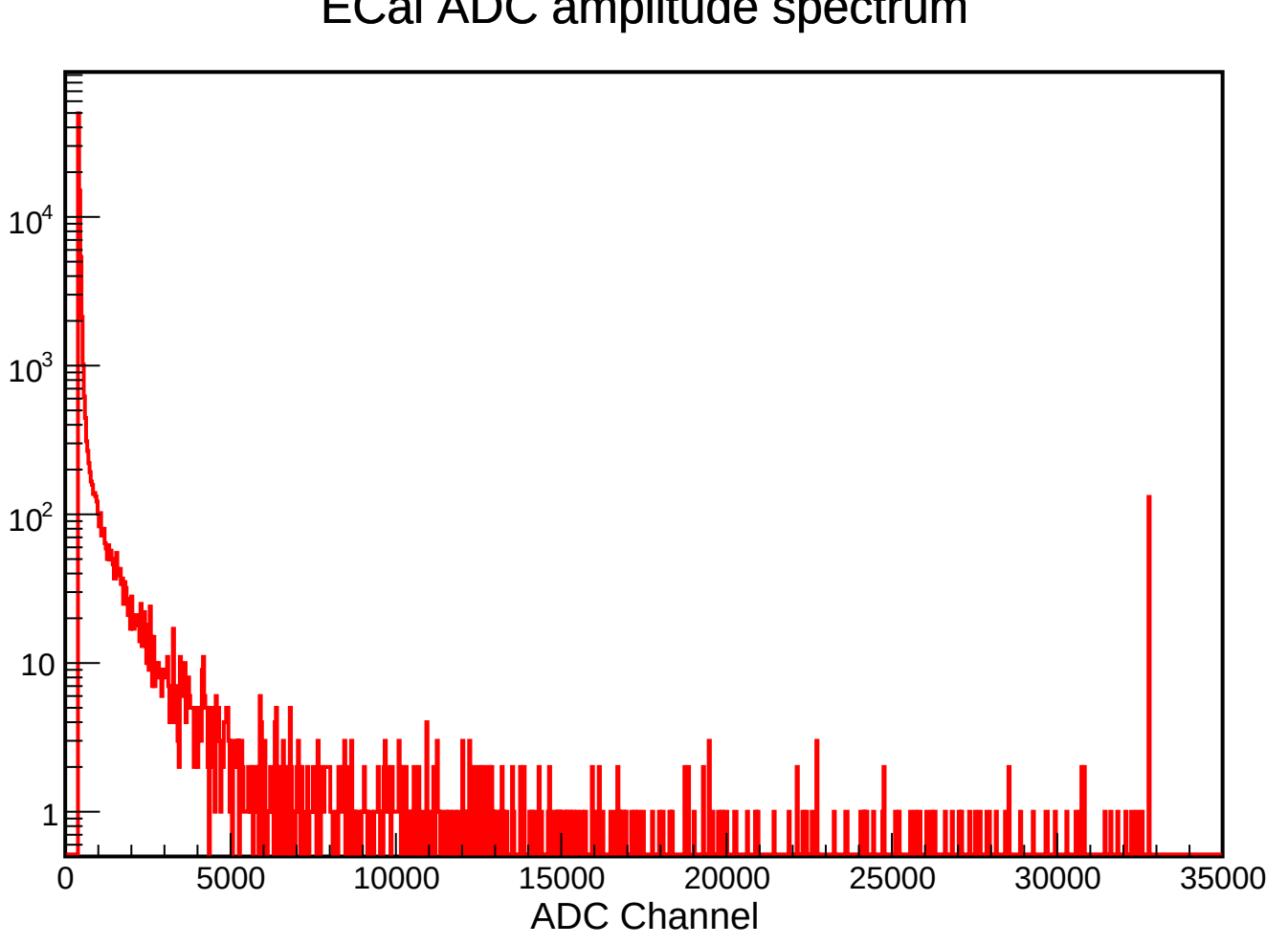
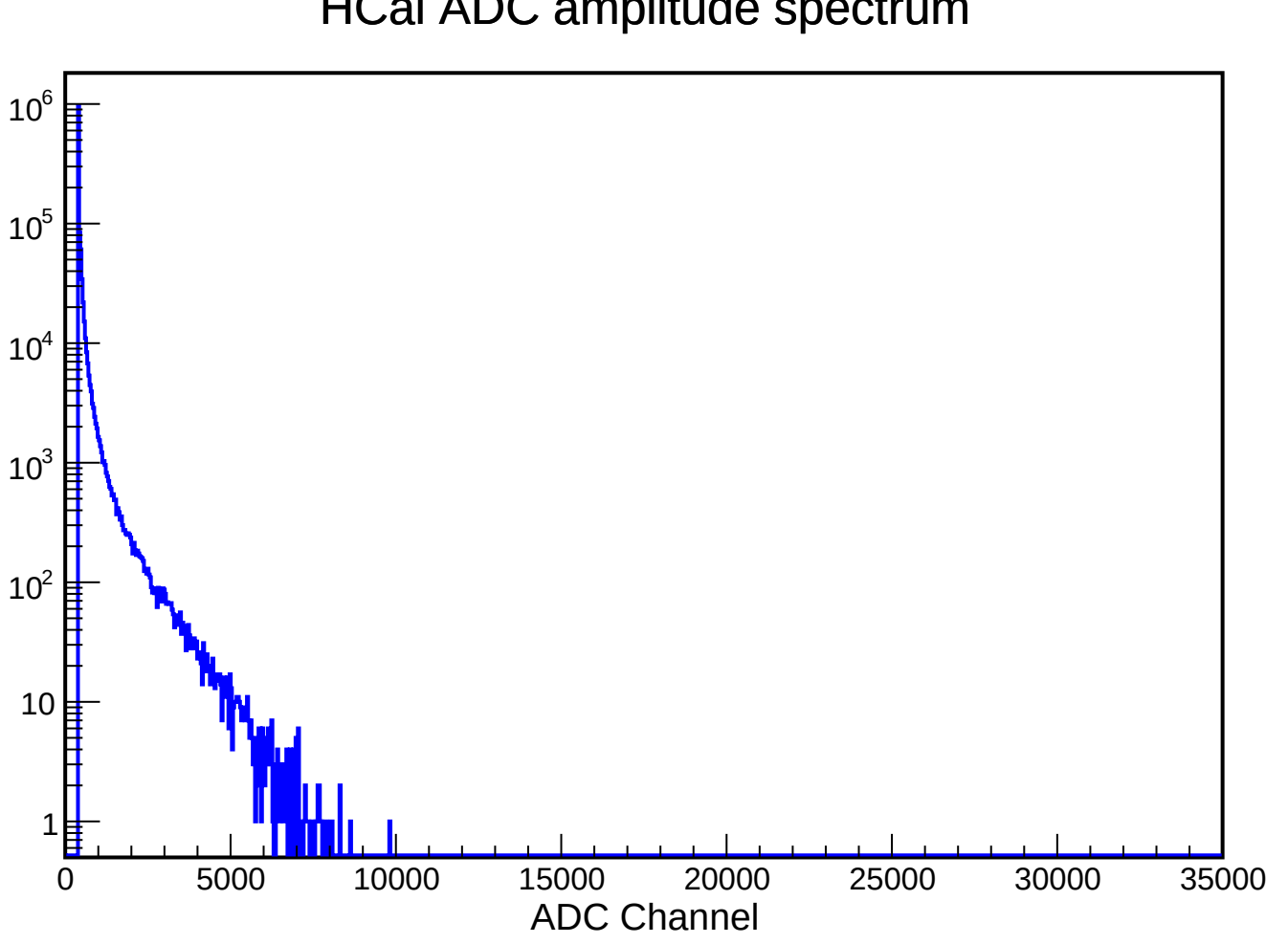

Neutron true energy vs. polar angle
E [GeV]
θ [deg]
0
20
40
60
80
100
120
140
160
180
200
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2
0
20
40
60
80
100

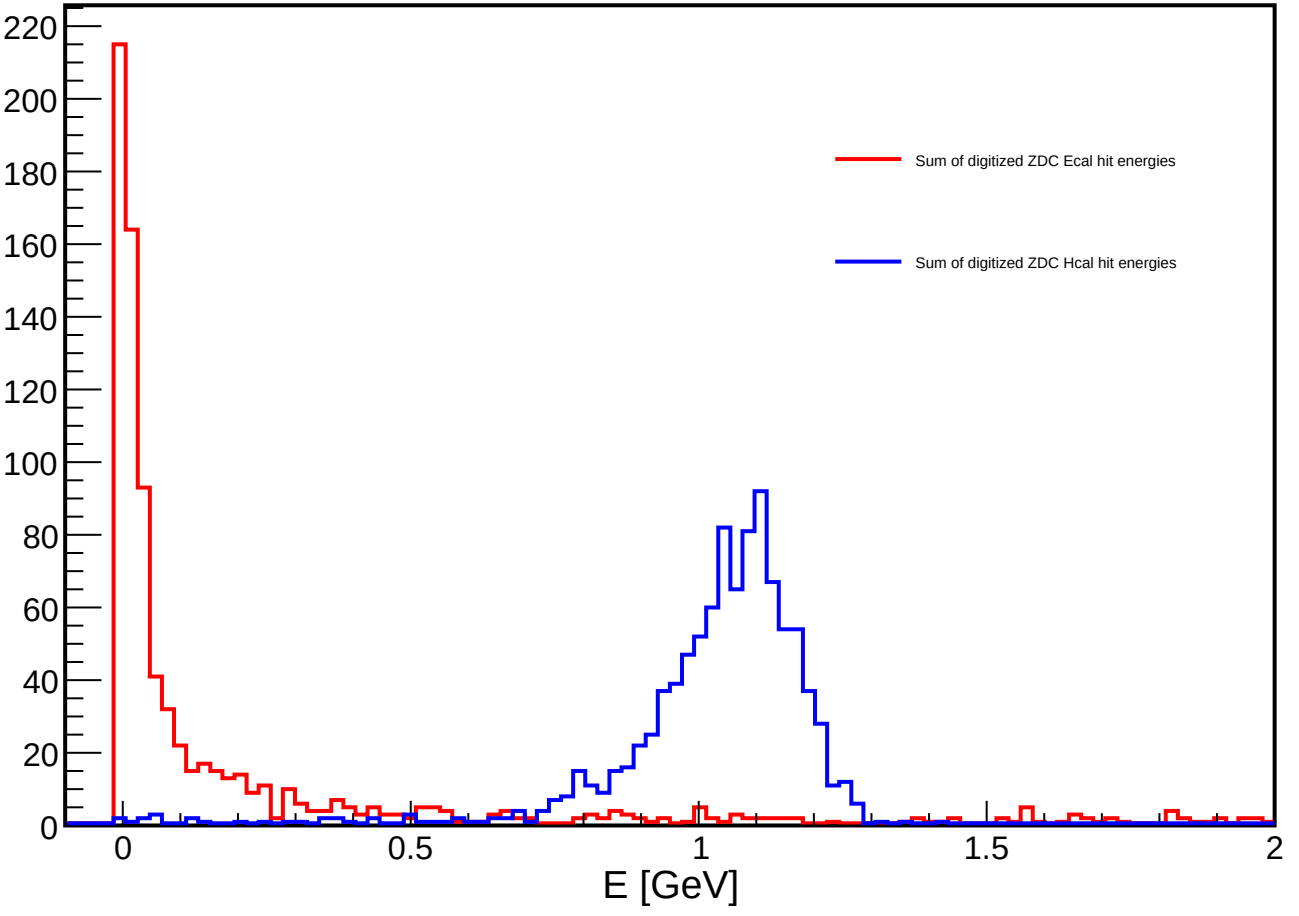

Neutron true azimuthal angle vs. polar angle around p axis
200
150
100
50
0
−50
−100
−150
−200
ϕ^* [deg]
0
2
4
6
8
10
12
θ^* [mRad]
4
3.5
3
2.5
2
1.5
1
0.5
0


Neutron true azimuthal angle vs. cosine of polar angle around p axis
200
150
100
50
0
−50
−100
−150
−200
ϕ^* [deg]
0.99996
0.999965
0.99997
0.999975
0.99998
0.999985
0.99999
0.999995
1
$\cos(\theta^*)$
6
5
4
3
2
1
0

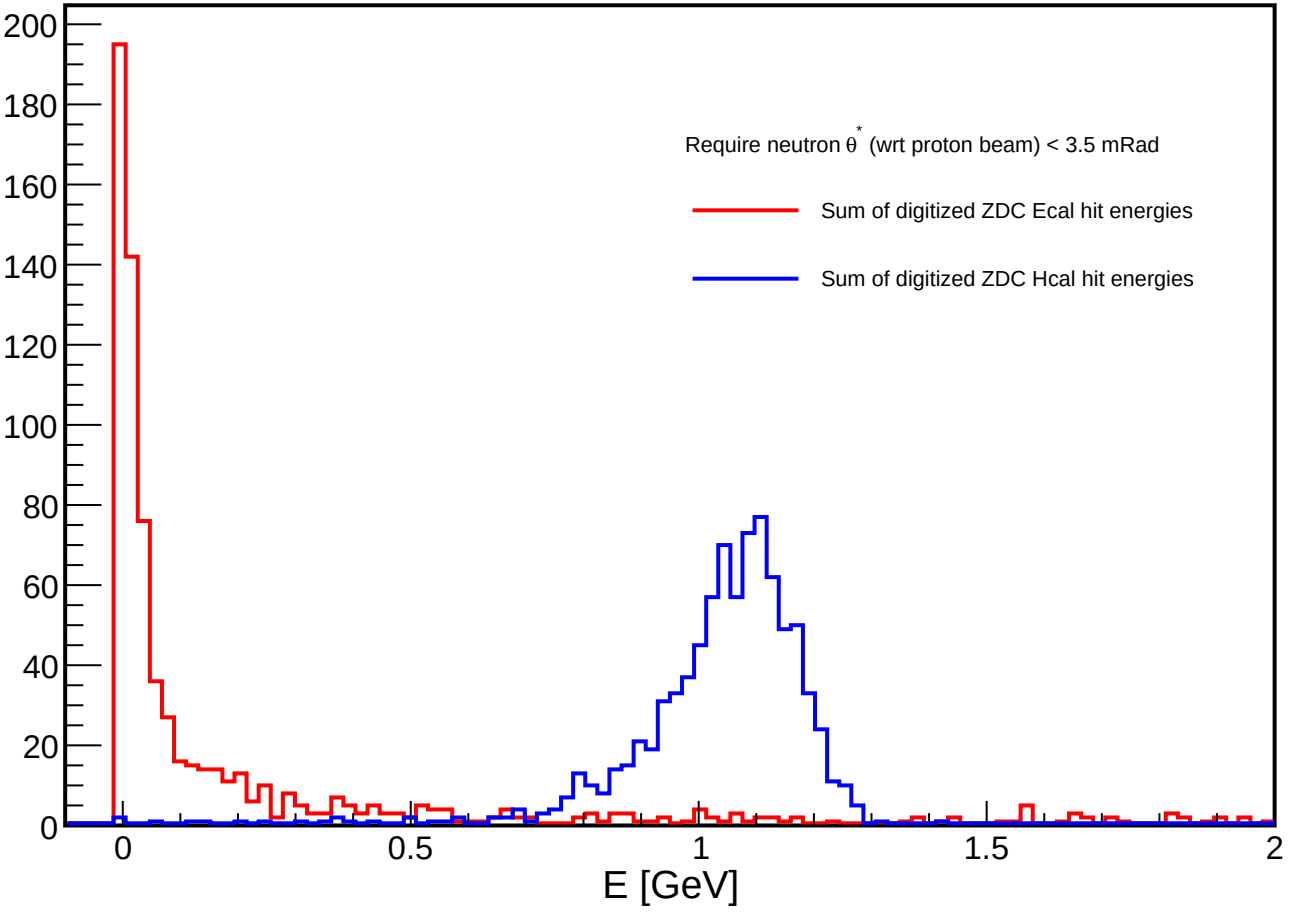

ECal ADC amplitude spectrum
10^4
10^3
10^2
10
1
0
5000
10000
15000
20000
25000
30000
35000
ADC Channel

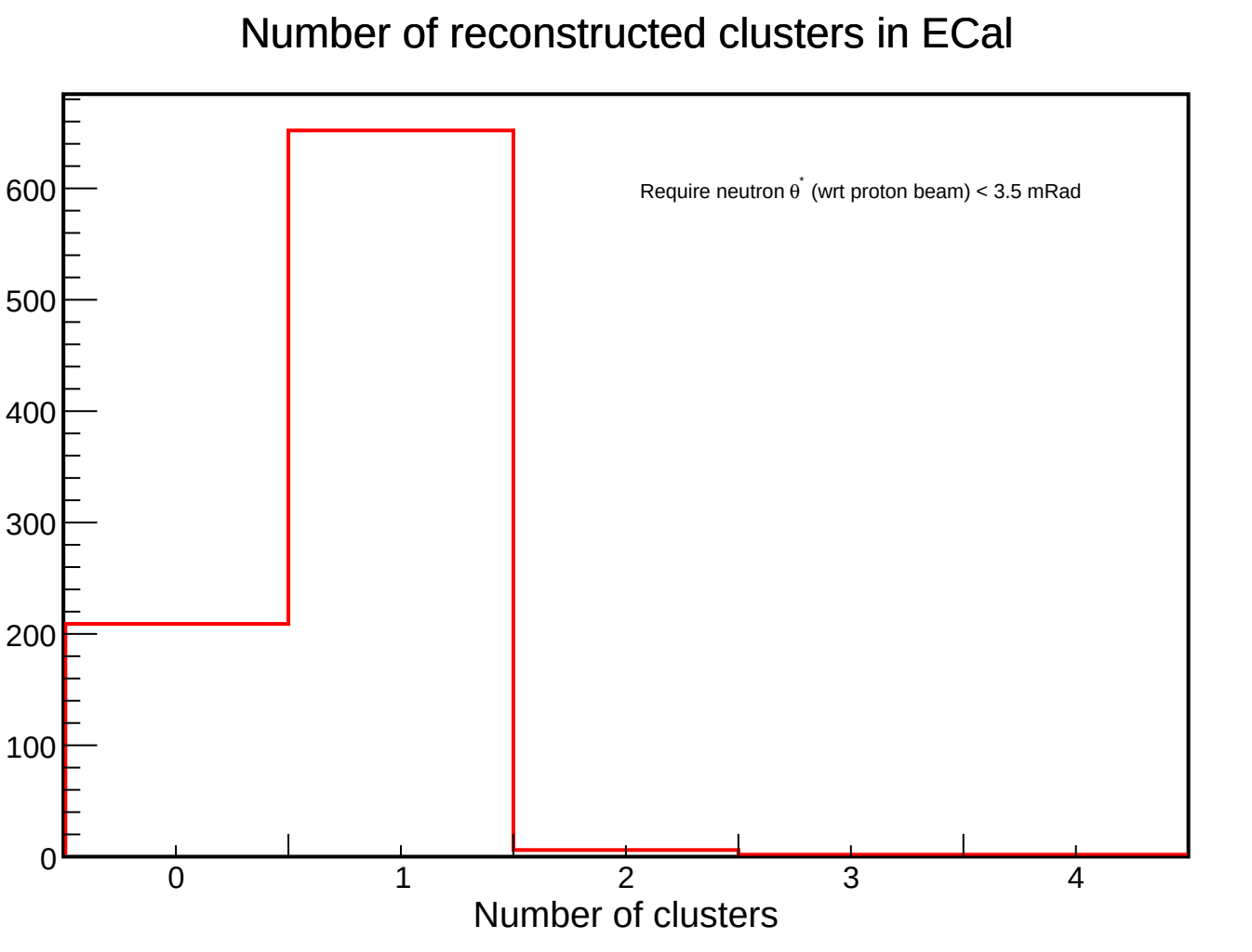

HCal ADC amplitude spectrum
10^6
10^5
10^4
10^3
10^2
10
1
0
5000
10000
15000
20000
25000
30000
35000
ADC Channel

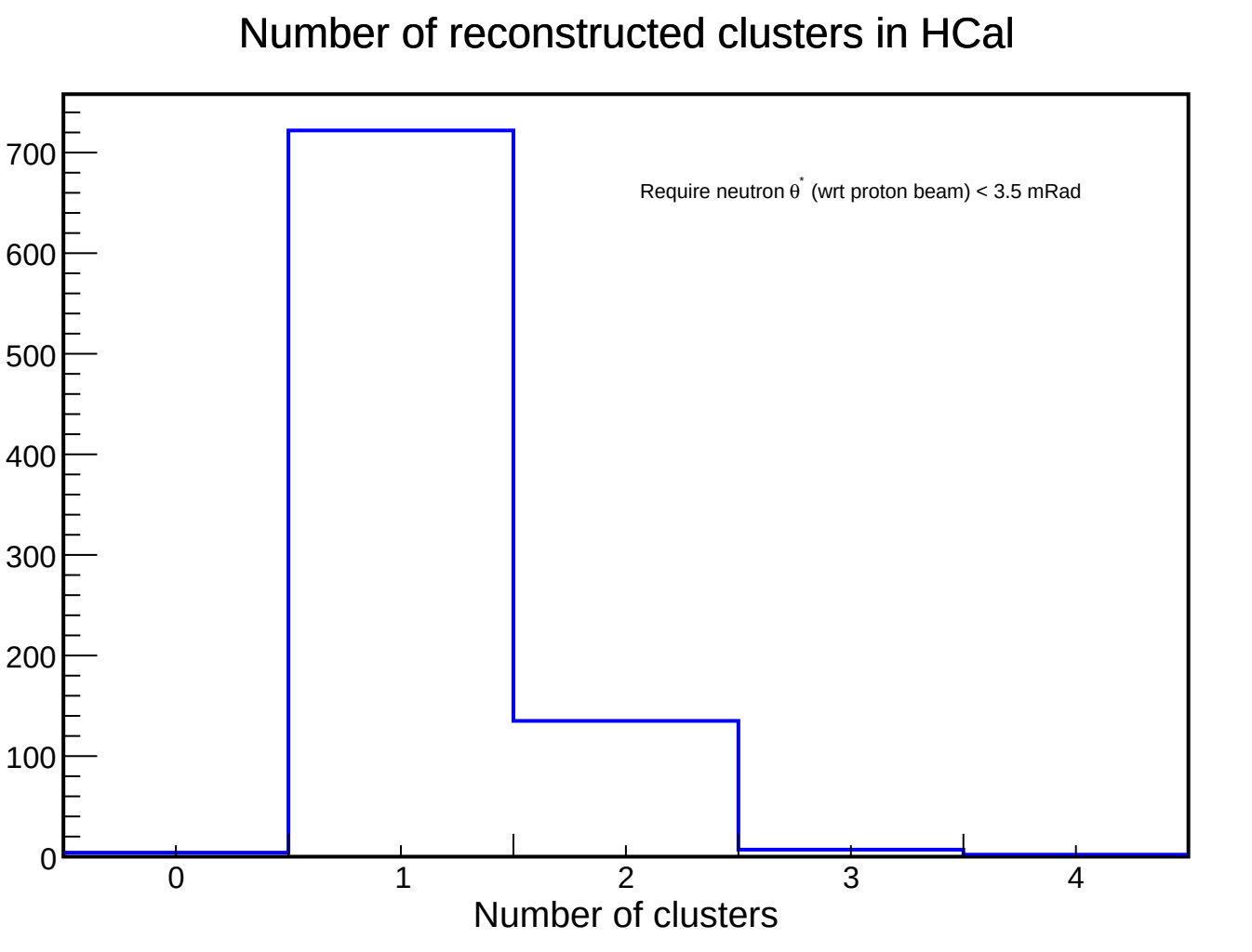
Total reconstructed hit energy sum

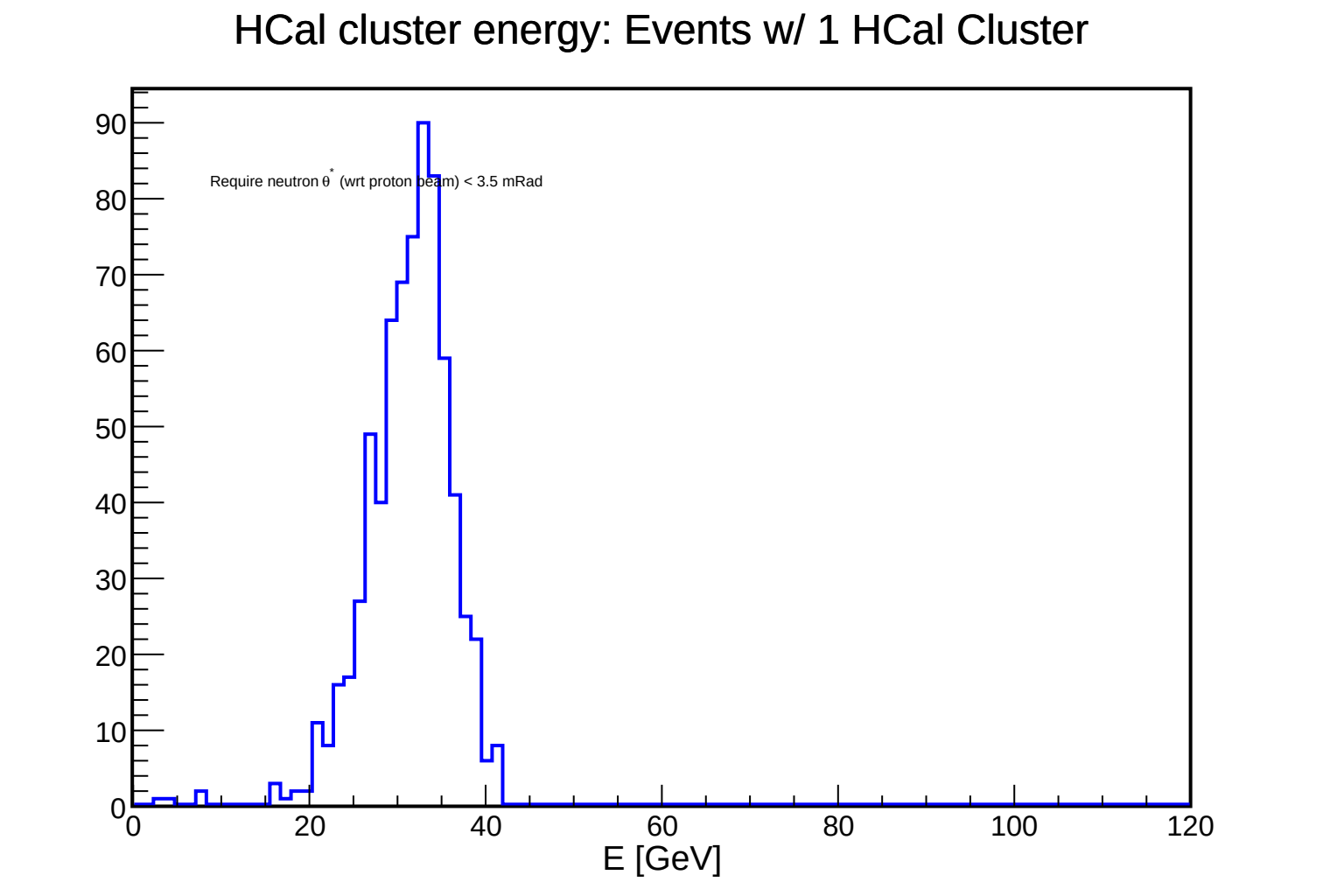


Total reconstructed hit energy sum

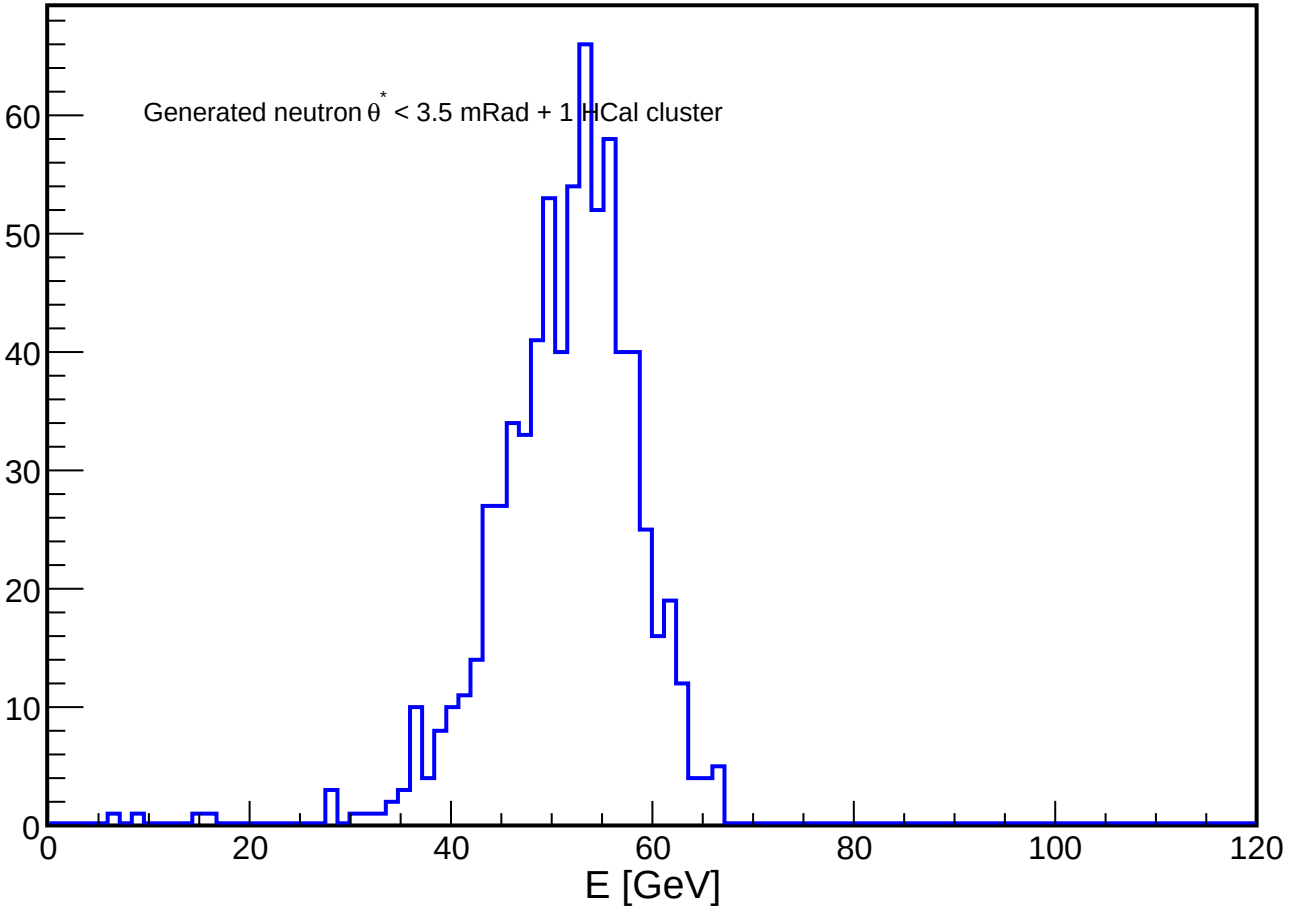


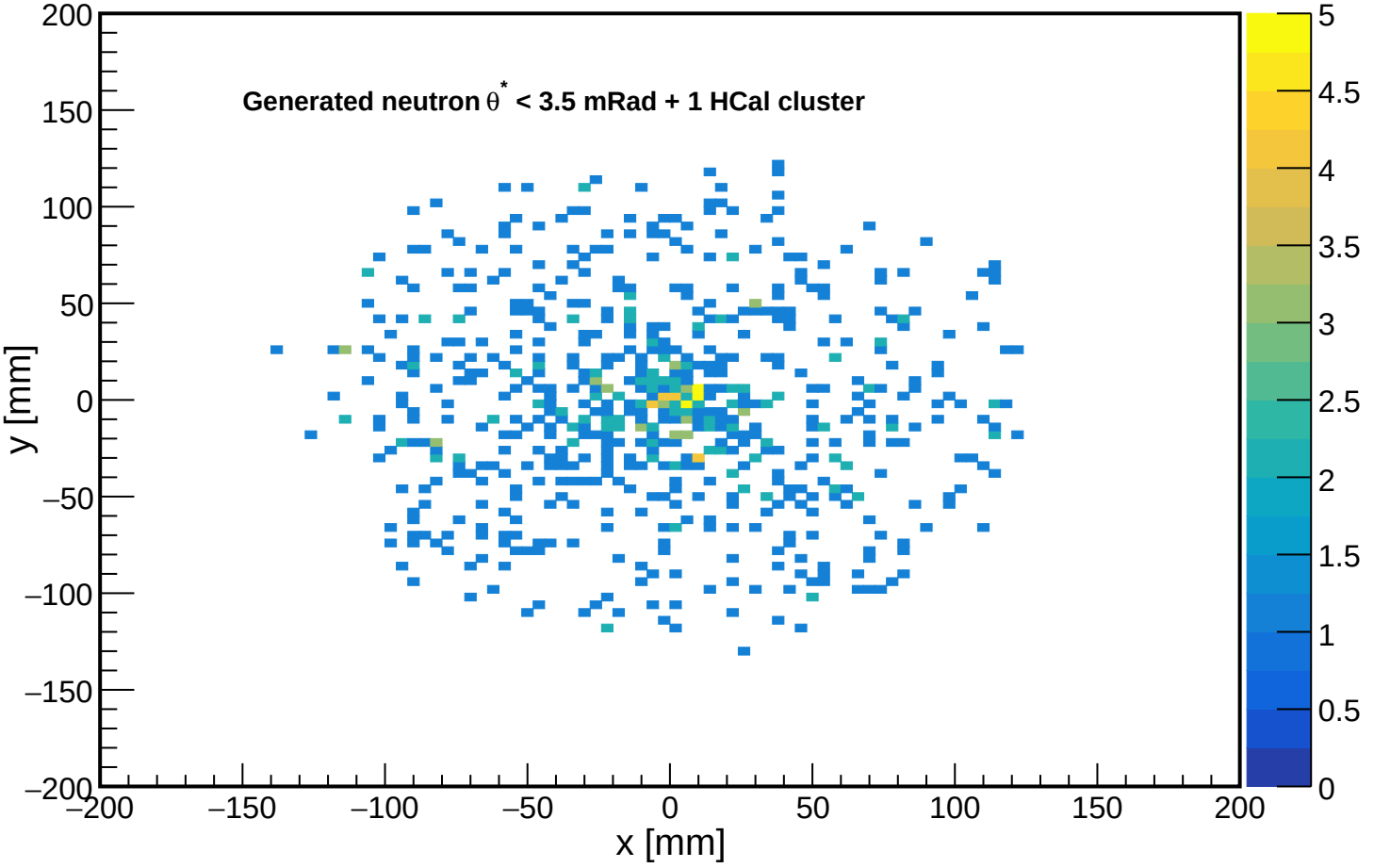

Number of reconstructed clusters in ECal
Require neutron θ* (wrt proton beam) < 3.5 mRad
600
500
400
300
200
100
0
0
1
2
3
4
Number of clusters


Number of reconstructed clusters in HCal
Require neutron θ* (wrt proton beam) < 3.5 mRad
700
600
500
400
300
200
100
0
0
1
2
3
4
Number of clusters


HCal cluster energy: Events w/ 1 HCal Cluster
Require neutron θ* (wrt proton beam) < 3.5 mRad
0
10
20
30
40
50
60
70
80
90
0
20
40
60
80
100
120
E [GeV]

Reconstructed Neutron Energy




Reconstructed Neutron local hit position at ZDC HCal front face
Generated neutron θ* < 3.5 mRad + 1 HCal cluster
y [mm]
x [mm]
200
150
100
50
0
−50
−100
−150
−200
−200
−150
−100
−50
0
50
100
150
200
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0